

Path-X: replacing the screening softmax with phase-coherent transport

The complex screening + PCR model introduced here is, to our knowledge, the first *genuinely complex-valued* neural network to solve Path-X: prior solvers are real-valued architectures that use complex numbers only inside a state-space kernel (S4, S4D, S5, LRU) or none at all (MEGA), whereas here the representation itself flows as a complex signal through both the screening and recurrence paths.

The one benchmark our 6-cell suite leaves open is **Path-X** (Long Range Arena; **Tay et al. 2021**): a 16,384-token binary connectivity task on which every softmax Transformer scores at chance, and which only complex-eigenvalue linear-recurrence models – S4, S4D, S5, LRU – and the EMA-plus-attention hybrid MEGA have solved. Path-X is *purely positional*: a 16k-length reach where selection alone is not enough. For the long-range positional information that `complex_screen` requires here, we replace its softmax with the phase-coherent recurrence (PCR) described below.

From softmax to phase-coherent recurrence (PCR). Rather than re-enabling the cosine softmax – a fixed, input-independent locality prior – we replace the positional slot of the screening stack with a **phase-coherent recurrence**: a complex diagonal linear recurrence $h_t = \lambda \odot h_{t-1} + \gamma \odot (Bx_t)$, applied bidirectionally by FFT convolution, interleaved with the `complex_screen` gate of §4. Following the LRU recipe (**Orvieto et al. 2023**) we use the stable parameterisation $\lambda = \exp(-\exp(\nu) + i\theta)$ with ring initialisation $|\lambda| \in [0.999, 0.9999]$, restricted phase $\theta \in [0, \pi/10]$, and γ -normalisation. This is the natural PCT choice of positional mechanism: the complex eigenvalue is an *input-independent phase rotation* per step – coherent long-range transport, the recurrence analogue of PCT’s multi-layer phase preservation (§5.1) – while the screening gate keeps its token-non-competing, real-valued selection. Screening now supplies selection; the phase-coherent recurrence supplies position. A cosine softmax is a static positional prior; the recurrence is a *learned, phase-coherent* one, and unlike the softmax it does not send screening to chance.

Result: a rule-compliant solve. Under the same rule-compliant setting as the leaderboard (raw 1D token sequence in, one binary label out; no 2D structure, no auxiliary supervision, no handcrafted features), the screening-plus-recurrence model solves Path-X, reaching **92.71 ± 0.89** test accuracy over three seeds (held-out split, deterministic sweep, $n=20,000$; best seed 93.50)¹ – between S4D-Inv and MEGA-chunk on the standard scale, and above S4-v1, DSS and S4D-LegS. Table 1 places this among the models that clear the 16k reach. Every softmax Transformer and efficient-attention variant (Reformer, Performer, Linformer, BigBird, Luna-256) scores at chance and is grouped in the first row; the solvers form a band from S4-v1’s 88.10 up to S5’s 98.58, with nothing established in between chance and the high 80s.

Model	Path-X
Transformer; Reformer, Performer, Linformer, BigBird, Luna-256	chance (≈ 50)
S4-v1	88.10
DSS	89.72
S4D-LegS	91.95
S4D-Inv	92.80
screen + phase-coherent recurrence (ours)	92.71 ± 0.89
MEGA-chunk	93.81
LRU	94.20
S4 (v2, S4-LegS)	96.35
MEGA	97.98
S5	98.58

Table 1: Rule-compliant Path-X test accuracy (%). Solvers ordered by score; all softmax Transformers and efficient-attention variants score at chance (first row). Our number is the mean over 3 seeds ($\pm$ sample s.d.; best seed 93.50), a first-pass run with no task-specific tuning. Baselines: S4-v1/S4-LegS and MEGA/MEGA-chunk from Ma et al. 2023; S4D-LegS/S4D-Inv and DSS from Gu et al. 2022 (S4D); S5 and LRU from Orvieto et al. 2023.

¹Model weights released on the Hugging Face Hub: <https://huggingface.co/complexedleo/pcr-screening-pathx>.

Two ablations mirror the framework. Both are single-variable and consistent with the PCT account. (i) *Phase necessity*. Fixing $\theta=0$ (real eigenvalues, an S4D-Real analogue, otherwise identical) never leaves chance – position without phase cannot form, the recurrence-side echo of anti-phase deletion breaking the cascade (§5.4). (ii) *Phase bandwidth is necessary for generalisation, not just for fitting*. A narrow initial phase band $\theta \in [0, \pi/50]$ drives training accuracy to 0.999 while test stays at 0.53 – perfect memorisation, no reach; widening to $[0, \pi/10]$ recovers genuine generalisation (0.92 test). Phase is thus required in a graded way: too little bandwidth and the model memorises, none at all and it cannot learn. This distinguishes “ignition” (train loss leaving $\ln 2$) from generalisation, which should be read only on held-out data.

Takeaway. The result is a first-pass run over three seeds with no task-specific tuning, and we do not claim to beat MEGA or S5. The claim is narrower and structural: the positional gap left by disabling the screening softmax is filled cleanly by a *phase-coherent* recurrence, and doing so clears the one long-range task no softmax Transformer solves – while the token-non-competing screening gate of §4 is retained unchanged for selection. The phase-coherence principle that governs the attention comparison of §4 thus extends, intact, to the positional mechanism of the same screening layer.

References (Path-X additions)

State-space models and long-range recurrence

- **Tay et al. 2021** – Yi Tay, Mostafa Dehghani, Samira Abnar, Yikang Shen, Dara Bahri, Philip Pham, Jinfeng Rao, Liu Yang, Sebastian Ruder, Donald Metzler. *Long Range Arena: A Benchmark for Efficient Transformers*. ICLR 2021. arXiv:2011.04006. <https://arxiv.org/abs/2011.04006>
- Defines the LRA suite including Path-X (16k-length Pathfinder); the rule-compliant setting used in §1.
- **Gu, Goel & Ré 2022** – Albert Gu, Karan Goel, Christopher Ré. *Efficiently Modeling Long Sequences with Structured State Spaces* (S4). ICLR 2022. arXiv:2111.00396. <https://arxiv.org/abs/2111.00396>
- First model to solve Path-X; established that a structured linear recurrence clears the 16k reach.
- **Gu et al. 2022** – Albert Gu, Ankit Gupta, Karan Goel, Christopher Ré. *On the Parameterization and Initialization of Diagonal State Space Models* (S4D). NeurIPS 2022. arXiv:2206.11893. <https://arxiv.org/abs/2206.11893>
- Diagonal SSM; its complex-vs-real ablation (S4D-Real fails Path-X) is the direct precedent for the phase-necessity ablation in §1. Source of the S4D-LegS (91.95), S4D-Inv (92.80), S4-v1 (88.10) and DSS (89.72) Path-X figures in Table 1 (Tables 5–6).
- **Gupta, Gu & Berant 2022** – Ankit Gupta, Albert Gu, Jonathan Berant. *Diagonal State Spaces are as Effective as Structured State Spaces* (DSS). NeurIPS 2022. arXiv:2203.14343. <https://arxiv.org/abs/2203.14343>
- The DSS baseline in Table 1; a complex diagonal SSM with softmax-normalised kernel, the first diagonal simplification of S4.
- **Smith, Warrington & Linderman 2023** – Jimmy T.H. Smith, Andrew Warrington, Scott W. Linderman. *Simplified State Space Layers for Sequence Modeling* (S5). ICLR 2023. arXiv:2208.04933. <https://arxiv.org/abs/2208.04933>
- **Orvieto et al. 2023** – Antonio Orvieto, Samuel L. Smith, Albert Gu, Anushan Fernando, Caglar Gulcehre, Razvan Pascanu, Soham De. *Resurrecting Recurrent Neural Networks for Long Sequences* (LRU). ICML 2023. arXiv:2303.06349. <https://arxiv.org/abs/2303.06349>

- Complex diagonal recurrence with ring init and γ -normalisation; states that restricted eigenvalue-phase initialisation is crucial for Path-X. The PCR recipe in §1 follows this parameterisation.
- **Ma et al. 2023** – Xuezhe Ma, Chunting Zhou, Xiang Kong, Junxian He, Liangke Gui, Graham Neubig, Jonathan May, Luke Zettlemoyer. *Mega: Moving Average Equipped Gated Attention*. ICLR 2023. arXiv:2209.10655. <https://arxiv.org/abs/2209.10655>
- EMA-plus-gated-attention hybrid; the strongest attention-adjacent Path-X baseline and the point of comparison for the token-competing vs. non-competing contrast.
- **Model weights (this work)** – Path-X screening + PCR checkpoint (Table 1), released on the Hugging Face Hub. <https://huggingface.co/complexedleo/pcr-screening-pathx>